

Joshua A. Bisdorf

joshua@bisdorf.org

linkedin.com/in/joshuabisdorf

github.com/joshuabisdorf

Ann Arbor, MI

bisdorf.org

Education

The University of Texas at Austin

Austin, TX

Master of Science, Computer Science

August 2026 – Expected May 2028

University of Michigan

Ann Arbor, MI

Bachelor of Science in Engineering, Computer Science | GPA: 3.67

August 2022 – December 2025

- Magna Cum Laude
- **Relevant Coursework:** Data Structures & Algorithms, Software Engineering, Computer Security, Cryptography, Machine Learning, Computer Vision, Computer Organization, Foundations of Computer Science, Data Science

Experience

KBR, Inc.

Ann Arbor, MI

Software Engineer | C++, MATLAB, Python

January 2026 – Present

- Engineered radar software for defense and intelligence systems, working across signal processing, machine learning pipelines, simulation, and computer vision
- Collaborated in an agile engineering environment to design, test, and integrate mission-critical software systems
- Streamlined data processing, software engineering, and algorithmic workflows to optimize radar application performance

Software Engineer Intern | MATLAB, Python

May 2025 – August 2025

- Refactored and migrated a Synthetic Aperture Radar (SAR) Automatic Target Recognition (ATR) framework from TensorFlow to MATLAB using the Deep Learning Toolbox, improving maintainability and streamlining deployment for engineering teams

Software Engineer Intern | MATLAB

May 2024 – August 2024

- Standardized and validated coordinate transformation utilities, including LLA, ECEF, ENU, and UTM conversions
- Developed image-processing tools to remove spurious artifacts and reconnect discontinuous regions in binary images
- Built visualization tooling for geospatial and radar-related objects in 3D Earth-space environments, including beam patterns, iso-range spheres, iso-Doppler cones, velocity vectors, and angular geometry

Harman International

Novi, MI

System Test Engineer Intern | Arduino, C#, HTML/CSS, PHP, SQL

May 2023 – August 2023

- Spearheaded the Harman Phone Management System, an internal platform for storing, securing, and managing engineering test devices
- Improved platform security by implementing HTTPS using OpenSSL-generated SSL certificates and securing authentication workflows through PHP session-based login management
- Integrated hardware and software systems by connecting an Arduino-controlled deadbolt to a web interface while developing administrative tooling for secure user and database management

Projects

Detecting Dirty Dish Culprits | Python, YOLOv8, Roboflow, OpenCV

- Built a YOLOv8-based computer vision pipeline to classify washing versus walking behavior in kitchen footage and flag likely dirty-dish drop-offs
- Annotated custom video data, added temporal continuity logic, and evaluated the detector with a 70/20/10 split, achieving 0.96 mAP50

Decrypting Blockchain: The Art of the Steal | Cryptography, Blockchain, Security Analysis

- Researched Bitcoin and Monero security/privacy tradeoffs, analyzing ring signatures, RingCT, stealth addresses, Dandelion++, traceability limits, and scam case studies

Bisdorf.org | Astro, TypeScript, CSS, Cloudflare Pages

- Built and deployed a responsive portfolio site with reusable Astro layouts, static project data, resume rendering/download support, project pages, and search indexing

SillyQL | C++, Hash Tables, Binary Search Trees, Query Processing

- Built a simplified SQL-style database engine using hash tables and binary search trees to support command-driven query processing and table joins

Network Security | Python, Network Security, Protocol Analysis, Packet Handling

- Implemented security-focused networking analyses with Python packet-handling tooling, evaluating protocol behavior, attacks, and defenses